AI ops assignment 1

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Question 1 — Technical Debt Diagnosis

1. Hidden Technical Debt Categories

(a) **Entanglement (CACE):** "The 14 undocumented shell scripts make the training pipeline difficult to understand and maintain because there is no proper orchestration."

(b) **Undeclared Consumers:** The marketing team is silently consuming the model's output table, creating an invisible dependency on the ML system.

(c) **Configuration & Glue-Code Debt:** The 14 undocumented shell scripts form a tangled pipeline without proper orchestration or a clear structure.

2. Mitigation

For (c), use a **DVC data pipeline** to explicitly define and organize the data-processing and training stages and their dependencies. This makes the pipeline more reproducible, organized, and easier to maintain.

Question 2 – MLflow Experiment Comparison

The best-performing run is mlp-lr-0.01-batch-128, with a validation accuracy of 0.949. It also has the lowest training loss (0.014) and a high training accuracy of 0.992. So, this combination of learning rate and batch size performed the best among the six runs.

There is some evidence of overfitting in the results. For example, the best run has a very low training loss of 0.014 and a training accuracy of 0.992, while its validation accuracy is 0.949. This means that the model performs better on the training data than on the unseen validation data. However, the difference is not very large, so the overfitting does not seem to be severe.

The learning rate appears to have a larger effect on performance than the batch size. When the learning rate changes from 0.001 to 0.01, the validation accuracy generally improves. For example, with batch size 128, the validation accuracy increases from 0.933 to 0.949. The batch size also affects the performance, but its effect seems smaller compared with the learning rate.